

JAX ML SCALING

BOOK

CHAPTER 1 AND 2

ROOFLINES

- FLOPS vs FLOPs
 - FLOPS: Floating point operations per second
 - FLOPs: Floating point operations
 - $\text{FLOPS} = \text{FLOPs/s}$
 - NVIDIA H100 can perform $9.89\text{e}14$ bf16 FLOPS

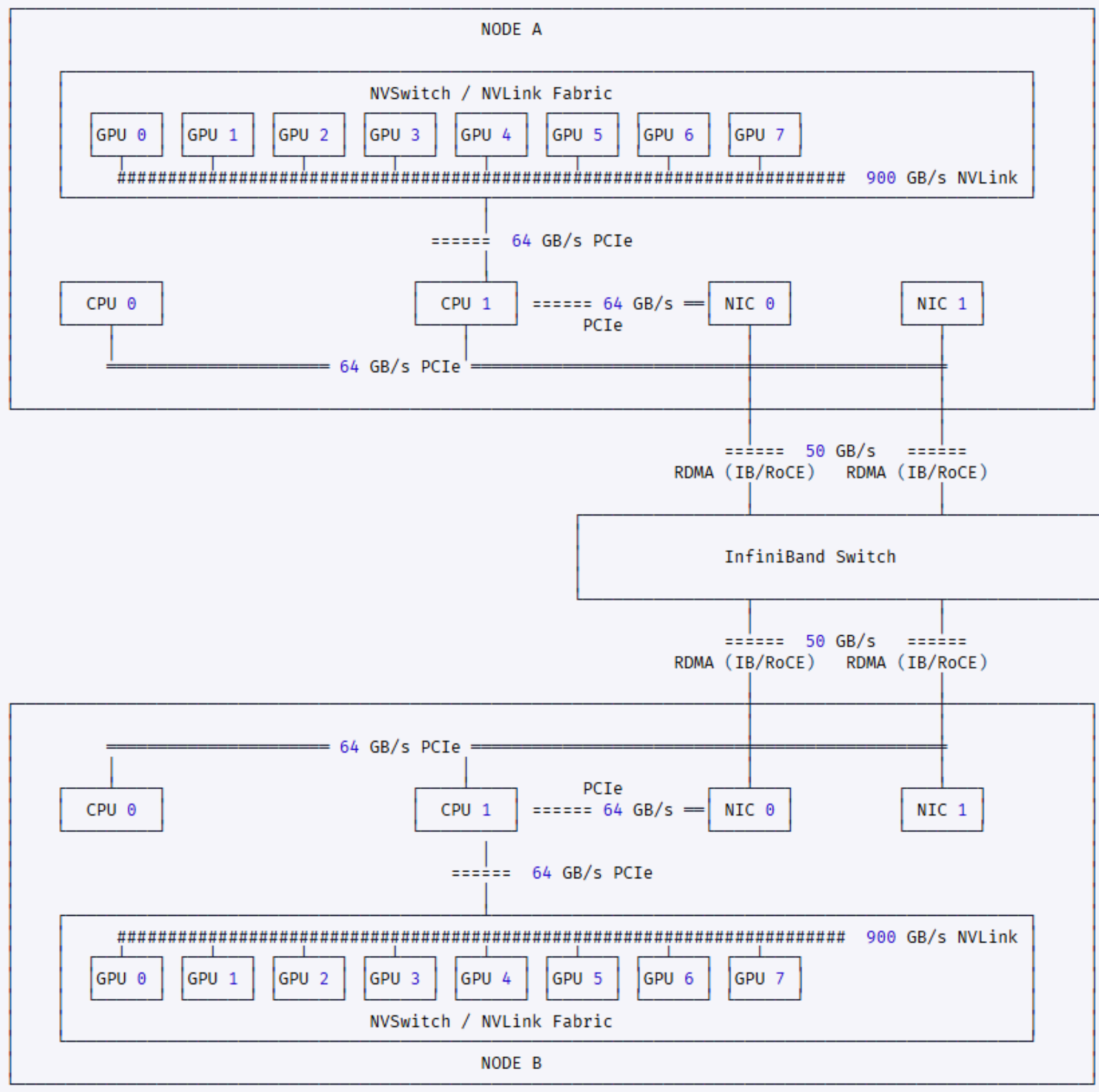
ROOFLINES

$$T_{math} = \frac{\textit{ComputationFLOPs}}{\textit{AcceleratorFLOPS}}$$

- NVIDIA H100 can perform 9.89e14 bf16 FLOPS
- To do 1e12 FLOPs
 - Time take = 1e12 / 9.98e14 = 1.1ms

$$T_{comm} = \frac{CommunicationBytes}{Bandwidth(Bytes/s)}$$

- The numerator is how many bytes we need to read and write
- Bandwidth comes from the connectivity in our GPUs or TPUs



Source

$$T_{math} + T_{comm} \leq 2 * (T_{math}, T_{comms})$$

$$T_{lower} = \max(T_{math}, T_{comms})$$

Overlapped math and communication
 Communication > math when we have to
 write the results back

$$T_{upper} = T_{math} + T_{comms}$$

Non-Overlapped math and communication

Arithmetic Intensity

- It is the ratio of the total FLOPs an accelerator performs to the number of bytes it needs to communicate
- We need to compare the arithmetic intensity of our algorithm and the accelerator to determine whether we are compute or memory bound

$$\text{Arithmetic Intensity} = \frac{\text{Computation FLOPs}}{\text{Communication Bytes}}$$

Example: Dot Product

a

b

c

d

e

f

$N = 3$

$$a * d + b * e + c * f$$

3 multiplications

2 additions (total 5)

$$N + N - 1 = 2N - 1$$

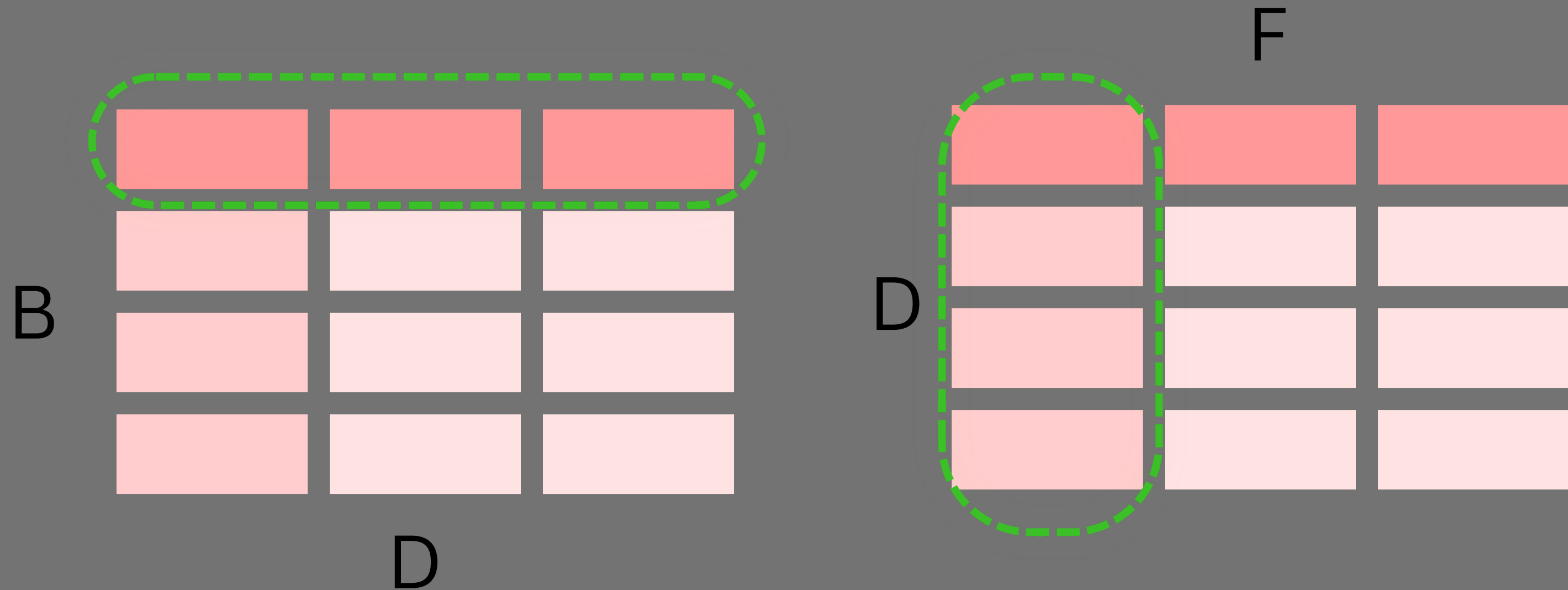
bf16 has 16 bits or 2 bytes

$$\text{Comms(bytes)} = 2N + 2N + 2$$

$$\frac{O(n)}{O(n)}$$

$$\text{Arithmetic Intensity} = \frac{2N - 1}{4N + 2} \approx \frac{1}{2}$$

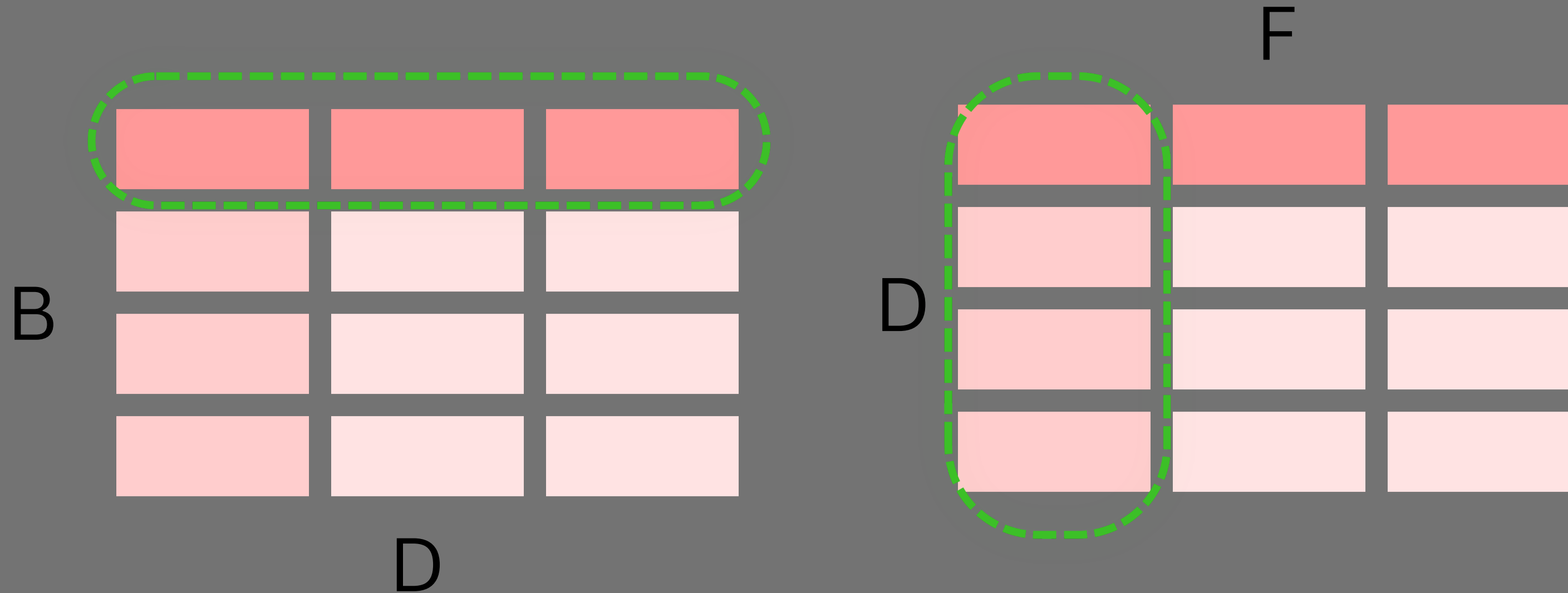
Matrix Multiplication



- D multiplications and $D - 1$ additions

$$(2D - 1)BF \approx 2BDF$$

Matrix Multiplication



$$\text{Read Bytes} = 2BD + 2DF$$

$$\text{Write Bytes} = 2BF$$

Arithmetic Intensity

$$\text{Intensity}(\text{matmul}) = \frac{2BDF}{2BD + 2DF + 2BF}$$

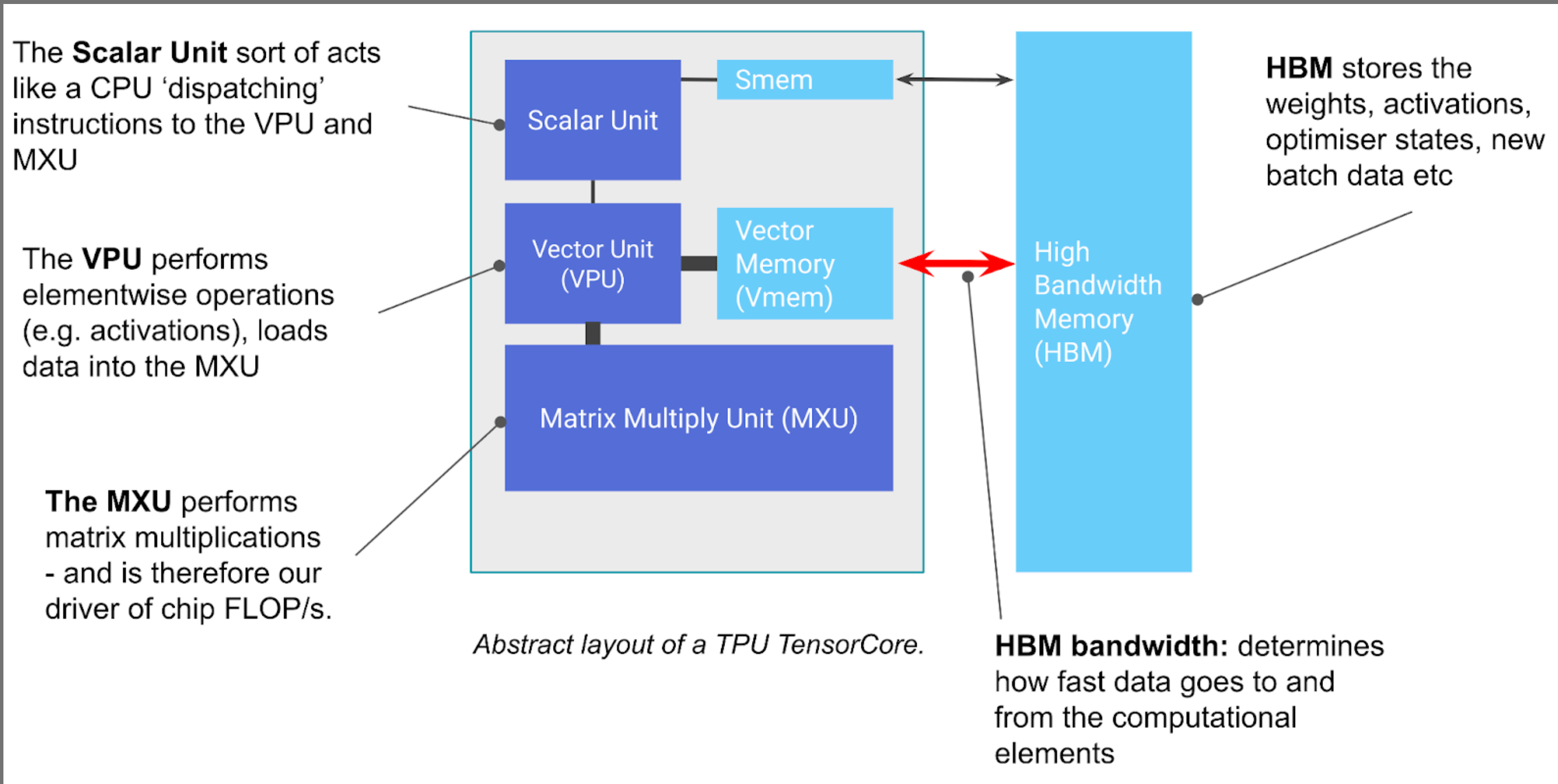
$$\frac{O(n^3)}{O(n^2)}$$

Work Problem

Question 4: What if we wanted to perform $\text{int8}[B, D] \cdot_D \text{int8}[B, D, F] \rightarrow \text{int8}[B, F]$ where we imagine having a different matrix for each batch element. What is the arithmetic intensity of this operation?

$$\text{Intensity} = \frac{2BDF}{BD + BDF + BF} \approx 2$$

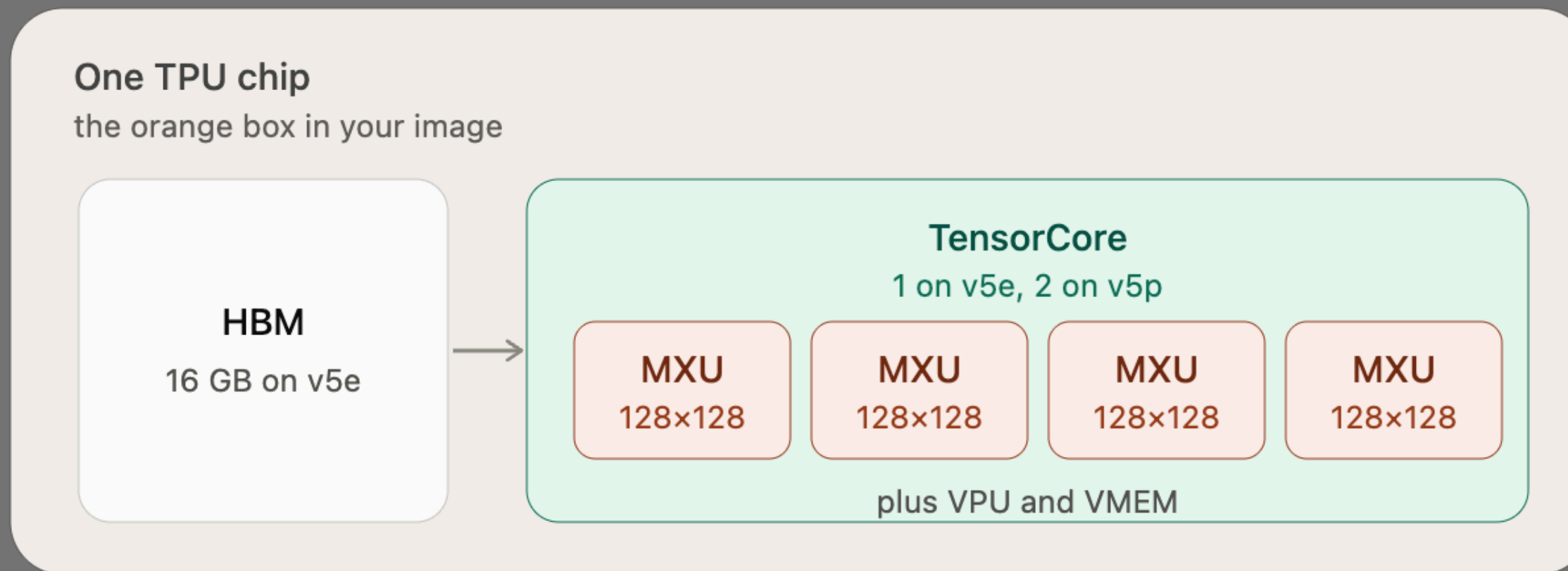
Chapter 2: TPU



Source

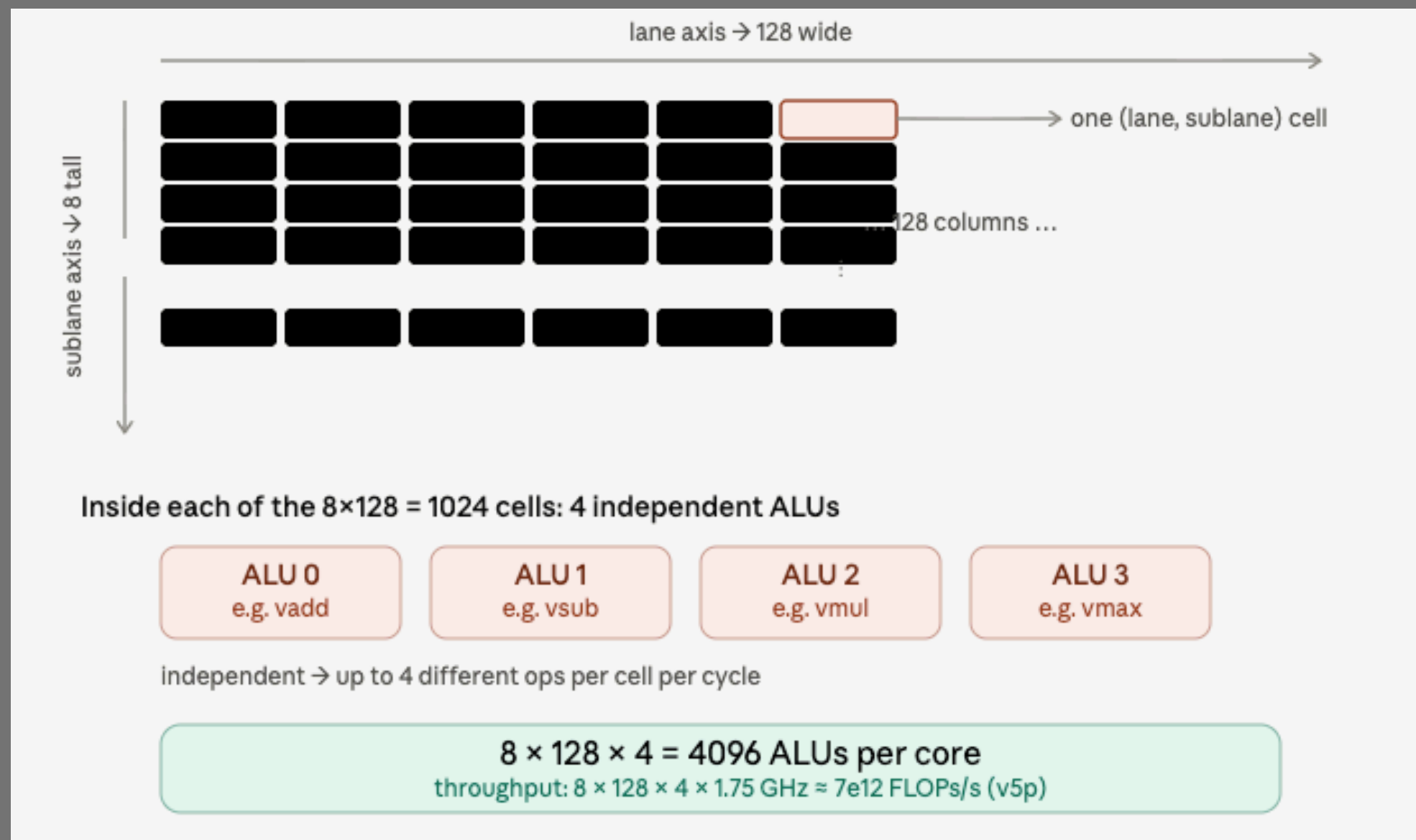
Components of TPU

- **MXU** → $bf16[8,128]$ @ $bf16[128,128]$ → $f32[8,128]$ matrix multiply₁ every 8 cycles using a systolic array. Instead of 128, it is 256 on TPU v6e
 - Most TensorCores have 2 or 4 MXUs, so e.g. the total bf16 FLOPs/s for TPU v5e is **2e14**. TPU v5e chip can do 4e14 int8 OPs/s



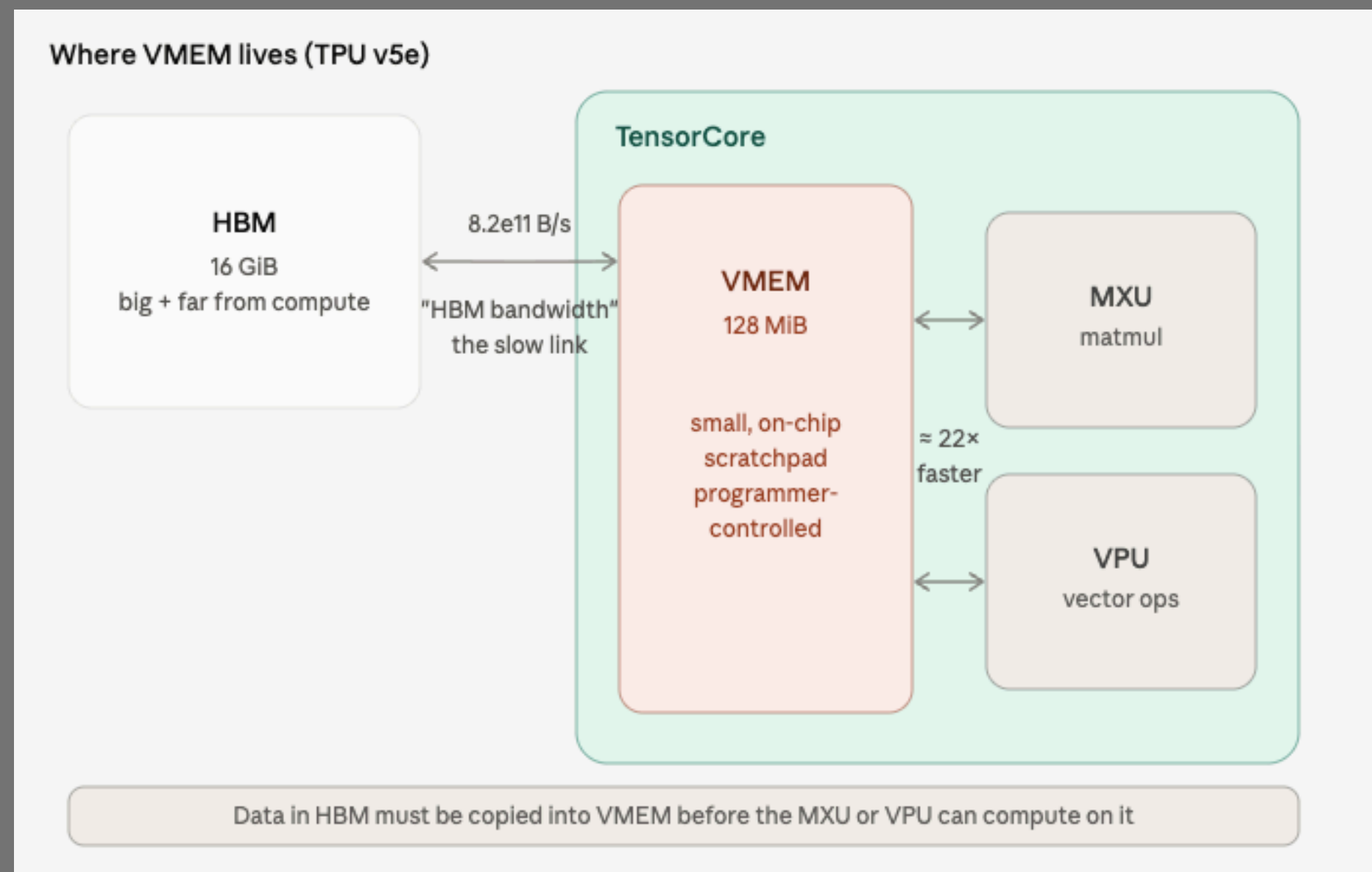
Components of TPU

- **VPU** (Vector Processing Unit) performs general mathematical operations like ReLU activations or pointwise addition or multiplication between vectors.



Components of TPU

- **VMEM** (Vector Memory) is an on-chip scratchpad located in the TensorCore, close to the compute units.



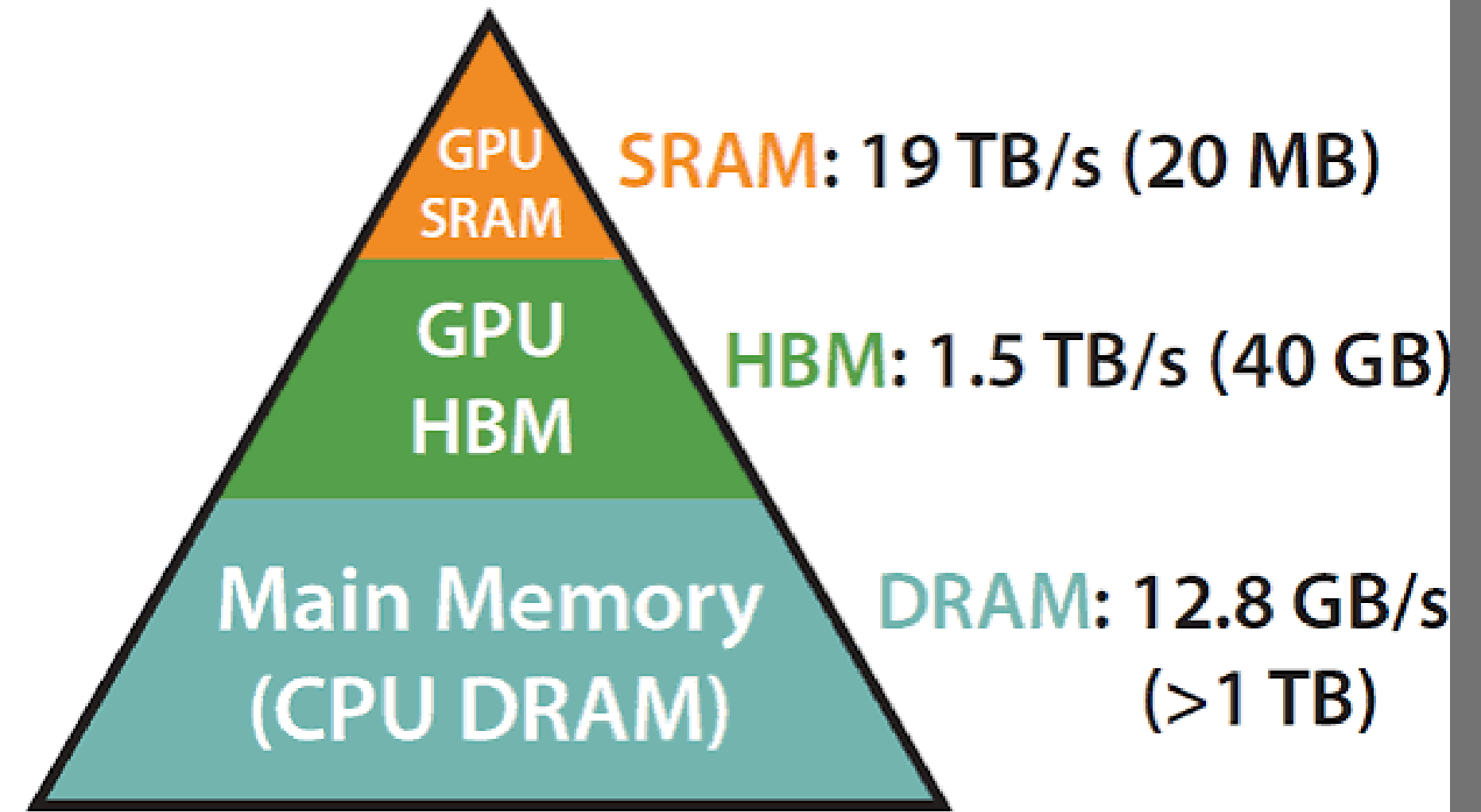
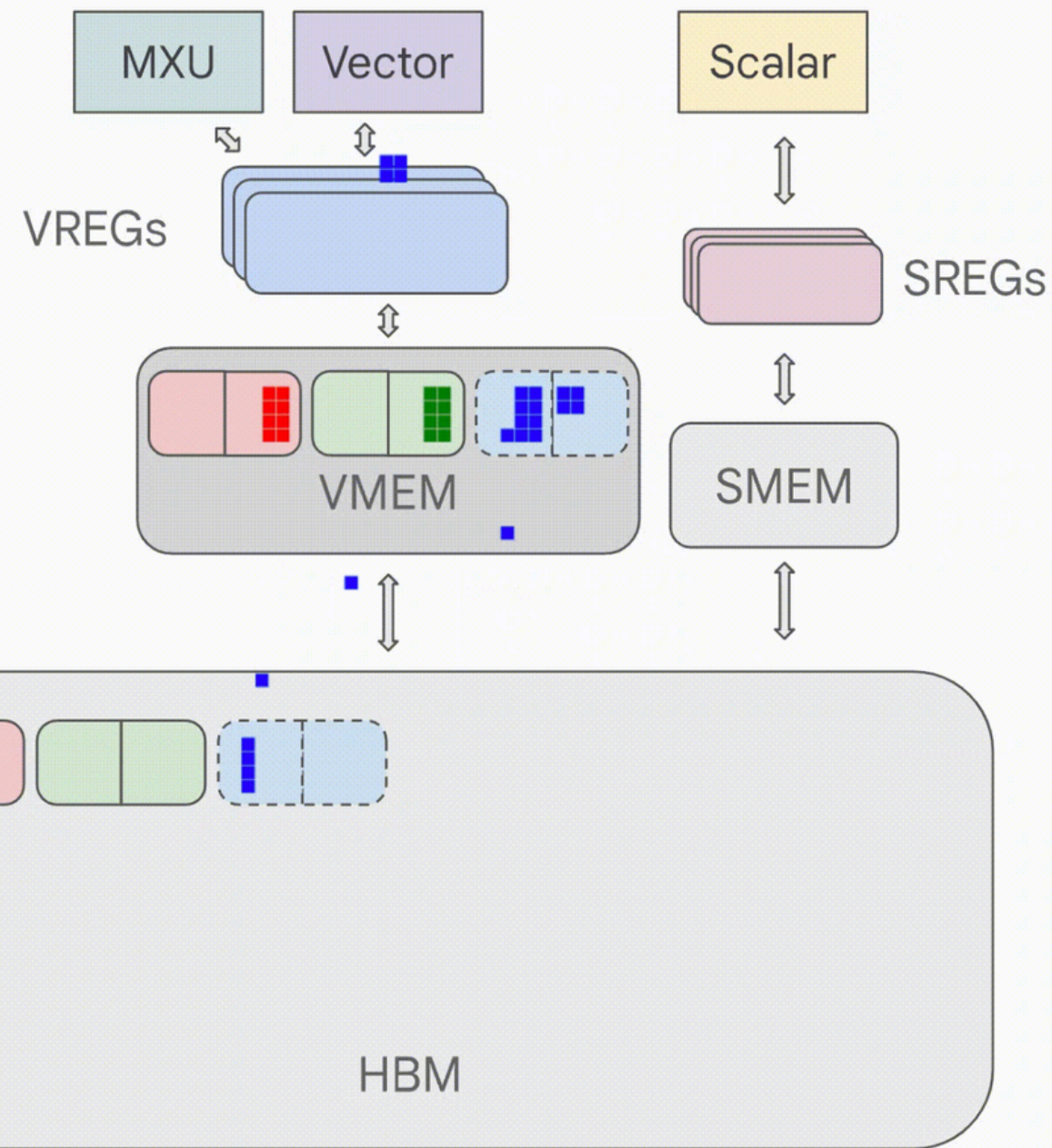
Components of TPU

- VREGs: Each TPU v5p core has 64 32-bit VREGs (32 in TPU v4), giving us a total of about $64 * 8 * 128 * 4 = 256\text{kB}$ of VREG memory per core

TPU

vs.

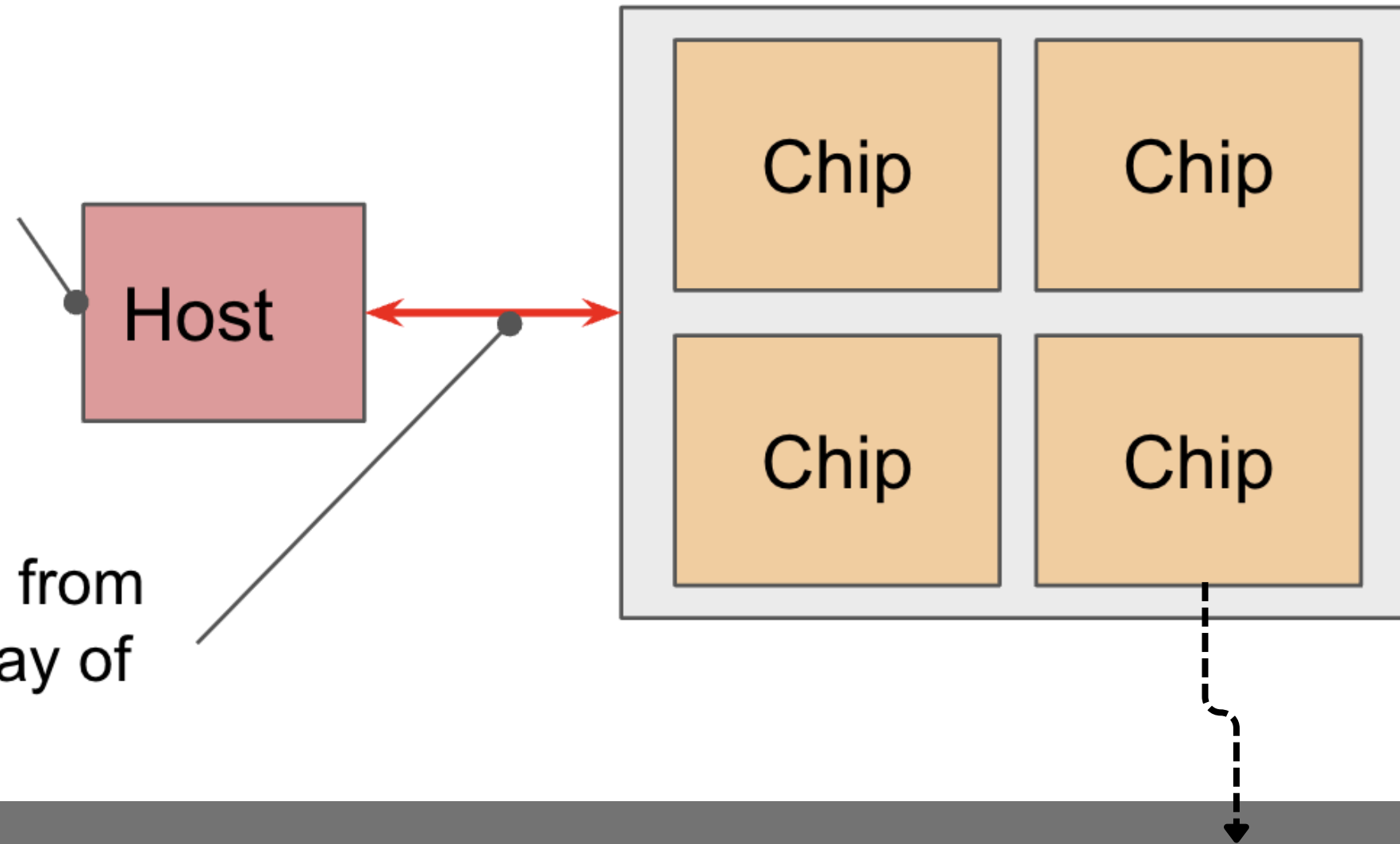
GPU



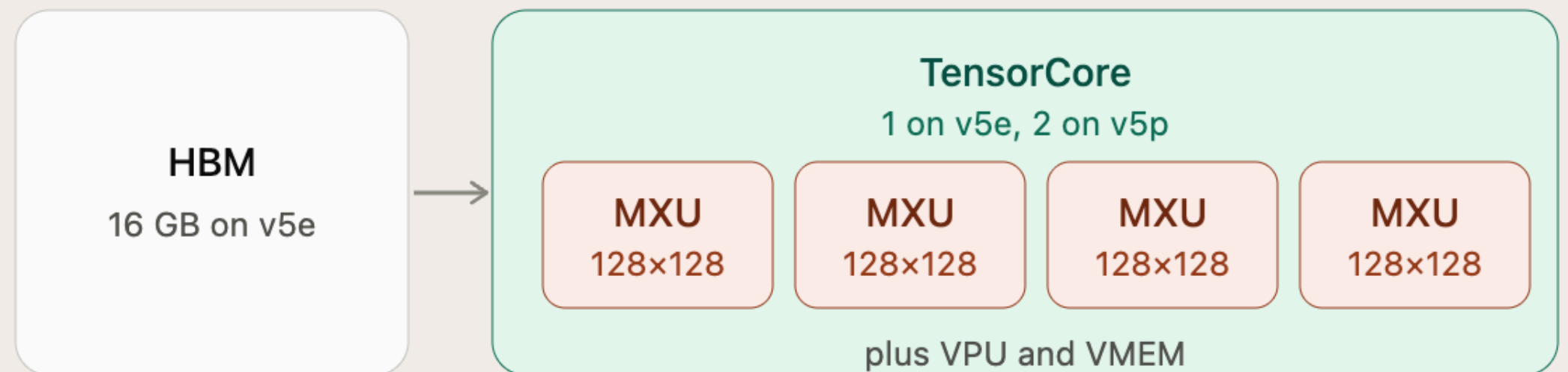
**Memory Hierarchy with
Bandwidth & Memory Size**

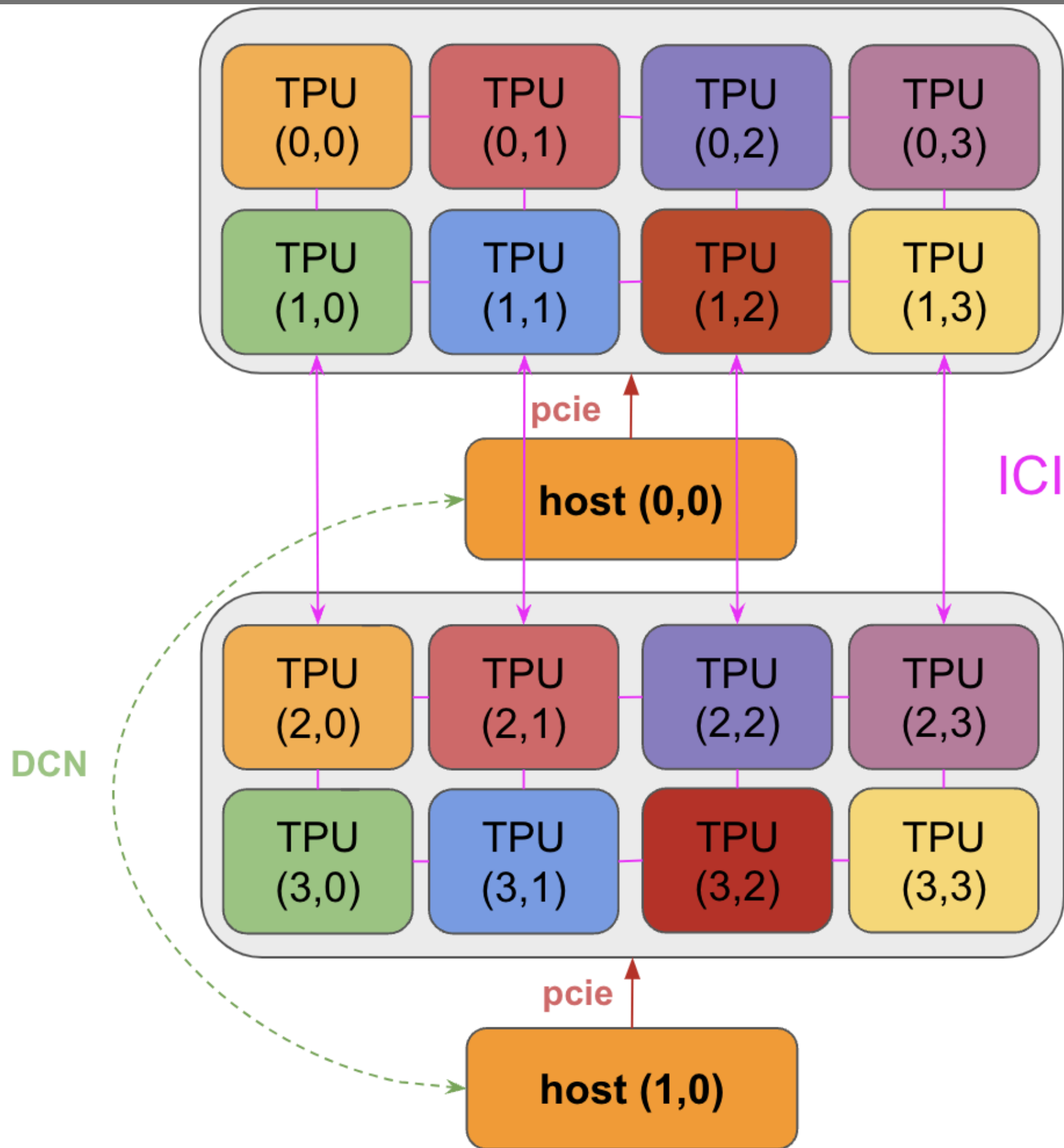
A host CPU which loads data, executes programs etc

PCIe connection from the host to the tray of chips



One TPU chip
the orange box in your image





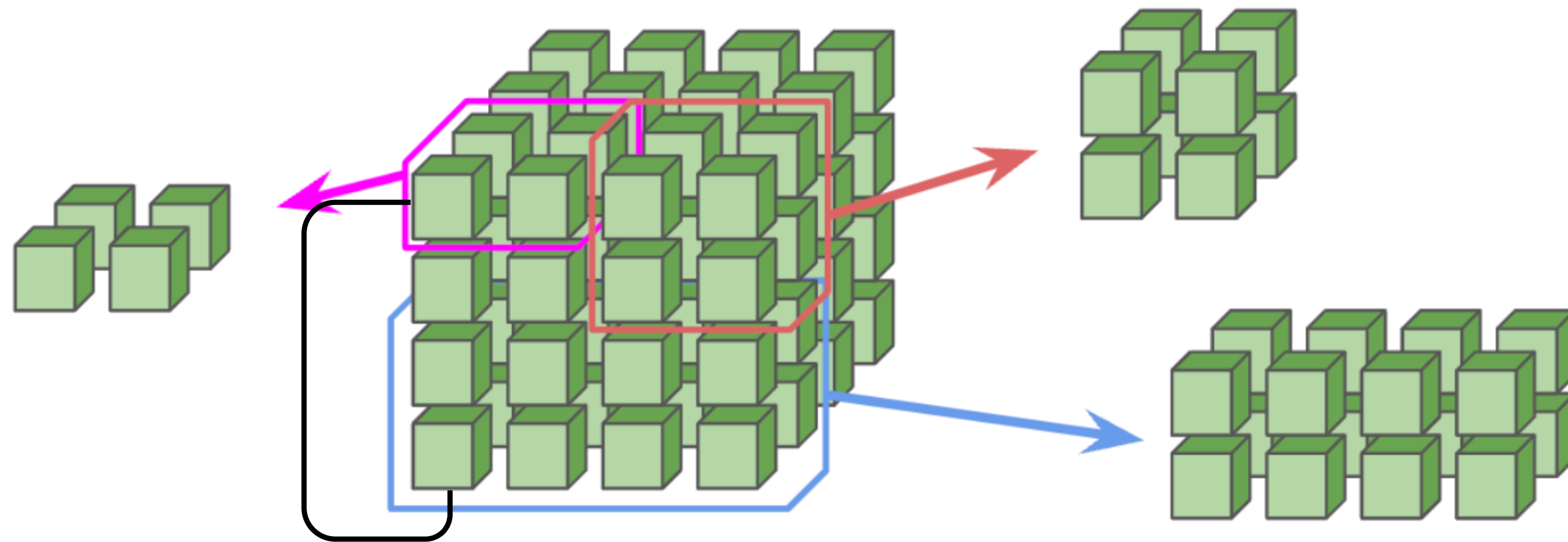
Source

TPU v5e

HBM > ICI > DCN

- 2.8e12 bytes/s (2.8 TB/s) of HBM bandwidth per chip.
- 9e10 bytes/s (90 GB/s) of ICI bandwidth per axis, with 3 axes per chip.⁷
- 6.25e9 bytes/s (6.25 GB/s) of DCN (egress) bandwidth per TPU (via 1-2 NICs on each host).⁸

Requesting TPU slices



Smaller slices can also be requested.

- This is helpful to split the TPU into Device Mesh.
 - We can split it into slices for Data Parallelism, Tensor and Pipeline Parallelism

SOLVING QUESTIONS

Q1

Q1) 200B parameter model. (bf16)
= 2 bytes.

Total memory = 400 GB. (200 x 2)

32 TPUv4p b/w = 1.2×10^{12} .

$$32 \text{ TPUs} \& \text{ 400GB} = \frac{400 \times 10^9}{32} = 1.25 \times 10^{10} \text{ per tpu.}$$

$$\text{Time taken} = \frac{1.25 \times 10^{10}}{1.2 \times 10^{12}} = 10 \text{ ms.}$$

Q2

Q2) TPUv5e

Pod size = 16 x 16

Host size = 4 x 2

$$\# \text{CPU hosts} = \frac{16 \times 16}{4 \times 2} = 32$$

#TPU tensor cores = 256

Total FLOPs = $256 \times 2e^{14}$.

Total memory = $256 \times 16 = 4 \text{TB}$.

TPU v5p. Pod size = 16 x 20 x 28.

Host size = 2 x 2 x 2.

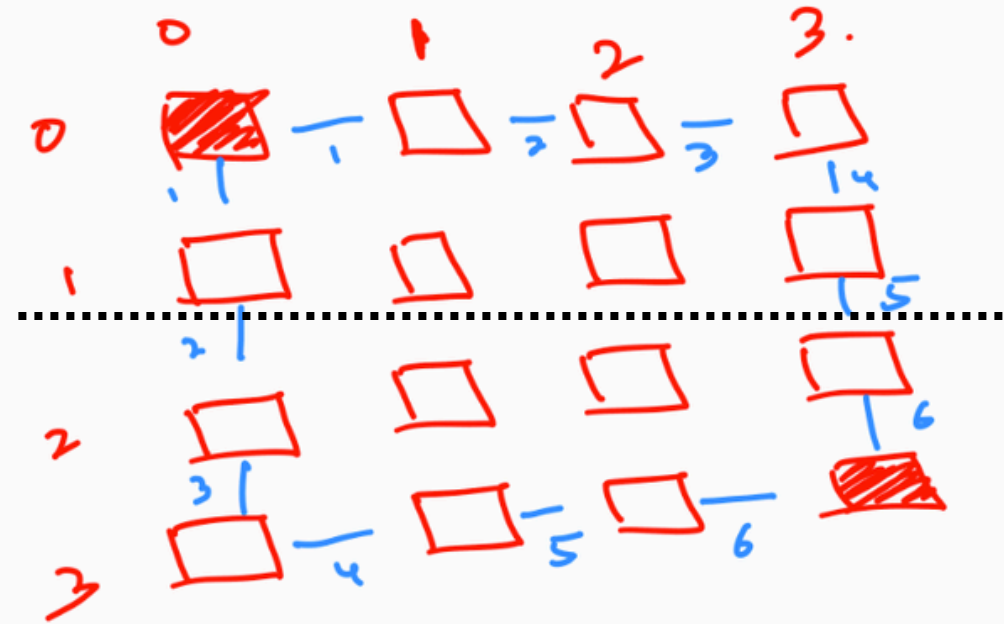
$$\# \text{CPU hosts} = \frac{16 \times 20 \times 28}{2 \times 2 \times 2} = 8 \times 10 \times 28 = 2240.$$

Total FLOPs = $16 \times 20 \times 28 \times 4.6e^{14} \times 2$

Total memory = $16 \times 20 \times 28 \times 96 \text{GB} \times 2$

Q3

Q5



Latency = 6 μ s.

$$B/w = 4.5 \times 10^{10}$$

$$2 \text{ direction} = 2 \times 4.5 \times 10^{10} = 9 \times 10^{10}$$

$$\begin{aligned} \text{total time} &= \frac{1.7 \times 10^7}{9 \times 10^{10}} = 188 \mu\text{s} + 6 \mu\text{s} \\ &= 194 \mu\text{s} \\ &+ \text{latency (6 } \mu\text{s)} \end{aligned}$$

$$16(8, 128, 3192)$$

$$= 2 \times 8 \times 128 \times 3192$$

$$= 1.7 \times 10^7$$